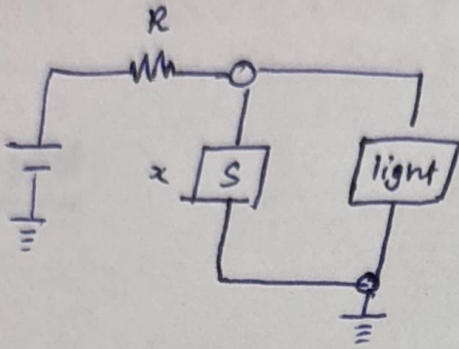


~~Series~~ SERIES $\rightarrow$ • AND
Parallel + OR

5-327

lec-2

NOT gate



why

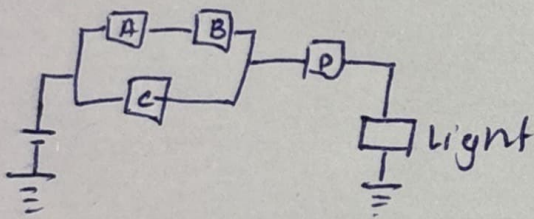
$$L(x) = \bar{x}$$

$$L = 1 \text{ if } x = 0$$

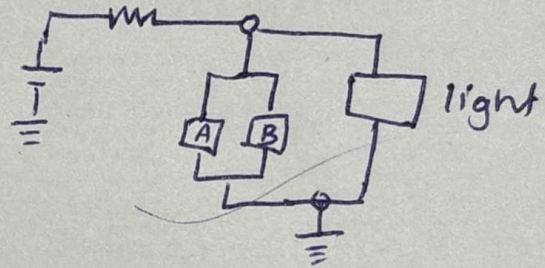
$$L = 0 \text{ if } x = 1$$

$$\bar{x} = x' = !x = \sim x$$

9) $((A \cdot B) + C) \cdot D$



$(A + B)'$



Binary number system

$$2^3 \ 2^2 \ 2^1 \ 2^0$$

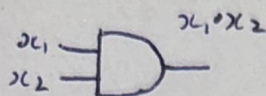
$$8 \ 4 \ 2 \ 1$$

$$1 \ 1 \ 1 = 7$$

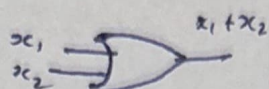
$$2^4 = 16$$

2^3	2^2	2^1	2^0	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

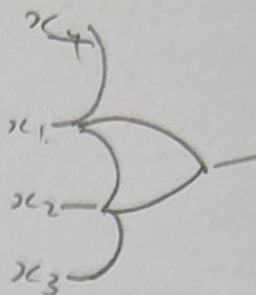
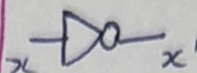
AND



OR

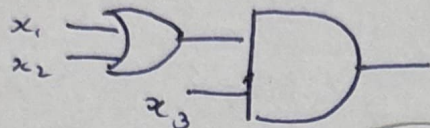


NOT



g) $(x_1 + x_2) \cdot x_3$

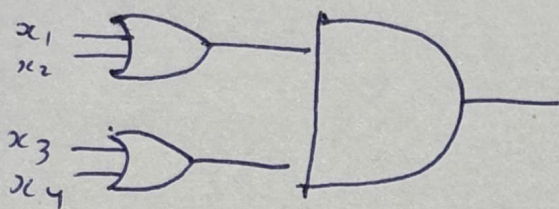
OR



$$2^3 = 8$$

x_1	x_2	x_3	$x_1 + x_2$	$\cdot x_3$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	0
1	1	1	1	1

g) $(x_1 + x_2) \cdot (x_3 + x_4)$



$$2^4 = 16$$

x_1	x_2	x_3	x_4	$x_1 + x_2$	$x_3 + x_4$	$(x_1 + x_2) \cdot (x_3 + x_4)$
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	0	0
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

Why parallel gives inversion

The parallel connection allows the switch to **bypass (short-circuit)** the lamp when it is closed.

- **Switch open:** current must pass through the lamp → lamp ON.
- **Switch closed:** current bypasses the lamp → lamp OFF.

This "bypass" effect is what creates the inversion.

Summary

- **Series connection:** switch controls whether current reaches the lamp → output follows input.
- **Parallel connection:** switch provides an easier path around the lamp → output becomes the opposite of the input.